

Bharath Kesav R

+91 7418297053 | bharathkesav1275@gmail.com |

 [Linked In](#) |  [GitHub](#) |  [PortFolio](#) |  [HuggingFace](#) |

Pollachi, Coimbatore, Tamil Nadu - 642114

OBJECTIVE

Integrated M.Sc. Data Science student at Amrita Vishwa Vidyapeetham, Coimbatore, with hands-on experience in Sports Analytics, Explainable AI, and Large Language Models. Built and deployed end-to-end ML pipelines across Football and Cricket match outcome prediction, IoT wearable sensor analytics, RAG-based LLM systems, and multi-task NLP — backed by 9 public GitHub repositories and 3 live deployed applications. Proficient in Python, PyTorch, scikit-learn, HuggingFace Transformers, SHAP, FAISS, SQL, Power BI, and Kaggle GPU-accelerated computing. Seeking an internship to apply data-driven solutions to real-world problems.

EDUCATION

• Amrita Vishwa Vidyapeetham

Int. M.Sc. Data Science

Sept 2022 - June 2027

Coimbatore, Tamil Nadu

- **Core Coursework:** Probability & Statistics, Linear Algebra, Calculus, Discrete Mathematics, Data Structures & Algorithms, Database Management Systems, Machine Learning, Deep Learning, Natural Language Processing, Big Data Analytics, Data Wrangling, and Computer Networks.
- **Electives:** Business Analytics, Statistical Quality Control, Soft Computing, GIS & Remote Sensing, Citizen Journalism

PROJECTS

• CricketGraph-DL: IPL Match Outcome Prediction & Player Impact Analysis

Jan 2026 - Mar 2026

Tools: PyTorch, PyTorch Geometric, SHAP, Optuna, IoT Simulation



- Designed a dual-branch architecture combining Graph Attention Network (GAT) over 652 IPL players with 7,334 batter-vs-bowler edges and BiLSTM temporal encoder fused via cross-attention Transformer for ball-by-ball win probability prediction
- Achieved test AUC-ROC of 0.7378 and Optuna-tuned validation accuracy of 69.33% on 225,954 deliveries across 950 matches; SHAP identified wickets fallen, wickets remaining, and balls remaining as dominant predictors

• Sarcasm Detection via Sentiment Incongruence: Multi-Task RoBERTa

Feb 2026 - Apr 2026

Tools: HuggingFace Transformers, PyTorch, LIME, scikit-learn, Pandas



- Proposed a novel Sentiment Incongruence Auto-Labeler deriving dataset-agnostic sarcasm labels from semantic mismatch between surface sentiment and underlying emotion using two parallel transformer pipelines
- Fine-tuned roberta-base with joint Focal Loss and WeightedRandomSampler achieving F1-Macro = 0.977 and AUC = 0.997 on 4,293 test samples; LIME token-level rationale extraction accompanies every prediction

• Explainable Match Summaries: RAG + LLaMA 3.1 + SHAP

Jan 2026 - Apr 2026

Tools: Sentence-BERT, FAISS, Groq API, SHAP, Pandas, Streamlit

 [[Live Demo](#)]

- Independently curated UCL-2025 dataset of 189 matches × 142 columns directly from uefa.com — deployed as a live RAG app achieving mean retrieval cosine similarity 0.903 vs 0.373 baseline
- Deployed LLaMA 3.1 via Groq API for hallucination-free grounded summary generation; SHAP LinearExplainer identified goal difference as the strongest match outcome predictor (mean |SHAP| = 1.000)

• Football Match Outcome Prediction Across European Leagues

Jan 2026 - Mar 2026

Tools: scikit-learn, SHAP, Pandas, NumPy, SQLite, Matplotlib, Seaborn

 [[Live Demo](#)]

- Built an end-to-end data pipeline engineering 33 statistical features from 25,979 historical match records across 5 European leagues — ELO ratings, dual rolling-window form, normalized betting probabilities from 3 bookmakers, and O(n) vectorized H2H win rates with a three-layer anti-leakage architecture
- Deployed live Streamlit app predicting outcomes for 259 teams across 5 leagues — Gradient Boosting outperformed ELO baseline by +0.073 F1-Macro with 0.91 xPts MAE

• Football Player Fatigue Prediction: Wearable IoT Sensors & ML

Nov 2025 - Jan 2026

Tools: scikit-learn, TensorFlow, Pandas, Matplotlib, imbalanced-learn



- Processed and analyzed 2.87M raw IoT sensor readings across 9 subjects using a leakage-free Karvonen heart rate labeling scheme; engineered 88 statistical features per sliding window (512 samples, 50% overlap) assigning three-class fatigue states (LOW/MEDIUM/HIGH)
- Personalized Random Forest achieved 97.87% accuracy and 79.93% F1-Macro; LOSO cross-validation confirmed 97.96%±2.57% accuracy across all subjects, exceeding the original PAMAP2 benchmark (~88%)

SKILLS

- **Programming Languages:** Python, SQL, LaTeX, HTML, CSS
- **Data Analytics & Visualization:** Microsoft Excel, Power BI, Tableau, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Statistical Analysis, Hypothesis Testing, Data Pipeline, Data Cleaning, Data Wrangling, EDA Dashboards, Data Storytelling, SHAP Plots
- **Machine Learning & Data Science:** scikit-learn, Feature Engineering, ELO Ratings, TimeSeriesSplit, SMOTE, LOSO Cross-Validation, Gradient Boosting, Random Forest, Logistic Regression, SVM
- **Deep Learning & NLP:** PyTorch, TensorFlow, HuggingFace Transformers, RoBERTa, T5, LSTM, BiLSTM, TCN, Graph Attention Networks (GAT), Multi-task Learning, Focal Loss
- **LLMs & Retrieval:** LLaMA, RAG, FAISS, Sentence-BERT, llama.cpp, Groq API, Prompt Engineering
- **Explainability (XAI):** SHAP (KernelExplainer, TreeExplainer, LinearExplainer, PermutationExplainer), LIME, Attention Visualization
- **Databases & Backend:** PostgreSQL, SQLite, Django, REST APIs, Streamlit
- **Tools & Platforms:** Git, GitHub, Kaggle (T4/P100 GPU), Google Colab, Jupyter Notebook, VS Code, Overleaf
- **Research Skills:** IEEE Paper Writing, Literature Review, Ablation Studies, Experimental Design, Cross-domain Evaluation, Scopus Citation

ACHIEVEMENTS

• Finalist — Pegasus 3.0 IEEE SB

Jan 2025

Muthoot Institute of Technology and Science (MITS)



- Selected as Finalist at Pegasus 3.0, an IEEE Student Branch technical competition organized by MITS
- Developed a Study Planner Guide powered by NLP Transformers to assist students with cognitive shortcomings including ADHD and dyslexia, enabling efficient task management and personalized learning support

ADDITIONAL INFORMATION

Languages: Tamil (Native), English (Professional), Malayalam (Basic)

Interests: Football, cricket, sports analytics, data analytics, Machine Learning & Deep Learning Research, open-source development, and sports AI.